

Exercice 4:

$$P(x) = (x+1)^3 (x^2 + \dots + 1)$$

Série 1: Poly et fct rationnelle

Exercice 1:

$$1) \quad 2x^4 - 3x^3 + 4x^2 - 2x + 6 \Bigg| \frac{x^2 - 3x + 1}{2x^2 + 3x + 11}$$

$$\boxed{28x - 5}$$

$$2x^4 - 3x^3 + 4x^2 - 2x + 6 = (2x^2 + 3x + 11)(x^2 - 3x + 1) + 28x - 5$$

$$2) \quad 2x + 3x^2 - x^3 \Bigg| \frac{1 + 2x - x^3}{2x - x^2 + x^3}$$

$$\boxed{-x^6 + x^6}$$

$$2x + 3x^2 - x^3 = (1 + 2x - x^3)(2x - x^2 + x^3) + (-x^6 + x^6)$$

$$= (1 + 2x - x^3)(2x - x^2 + x^3) + x^6(x - 1)$$

$$R(x) = x - 1$$

Exercice 2:

$$1) \quad P(x) = ax^{n+1} + bx^n + 1 \text{ soit divisible par } (x-1)^2$$

$$\Rightarrow P(x) = Q(x-1)^2 \Rightarrow 1 \text{ est une racine double de } P.$$

$$\Rightarrow \begin{cases} P(1) = 0 \\ P'(1) = 0 \end{cases}$$

$$P(1) = a + b + 1 = 0 \Rightarrow a = -b - 1$$

$$P'(x) = a(n+1)x^n + bn x^{n-1}$$

$$P'(1) = a(n+1) + bn$$

$$= (n+1)(-b-1) + bn$$

$$= -b(n+1) - (n+1) + bn$$

$$= -b - 1 = 0 \Rightarrow b = -1$$

$$\Rightarrow b = -(n+1)$$

$$\Rightarrow a = n+1-1 = n$$

$$(x^2+x+1) \Bigg| \frac{x^2+1}{x^2-1+1} = c_j + d$$

$$2) \quad P(x) = Q(x)(x+1)^2 + R(x)$$

$$\deg R < \deg (x+1)^2 = 2$$

$$\Rightarrow \deg R = 1$$

$$\Rightarrow R(x) = ax + b$$

$$P(x) = Q(x)(x+1)^2 + ax + b$$

(-1) est une racine double de P

$$P(-1) = b - a$$

$$P'(-1) = (-1)^n + n(-1)^{n-1} + a + 1 = b - a$$

$$P'(-1) = a$$

$$P'(x) = n x^{n-1} + n(n-1)x^{n-2} + 2x$$

$$P'(-1) = [n(-1)^{n-1} + n(n-1)(-1)^{n-2} - 2 = a]$$

$$\Rightarrow b = (-1)^n + n(-1)^{n-1} + 1 + n(-1)^{n-1} + n(n-1)(-1)^{n-2}$$

$$\boxed{b = (-1)^n + 2n(-1)^{n-1} + n(n-1)(-1)^{n-2}}$$

Exercice 3:

$$1) \quad x^4 + 1$$

dans $\mathbb{R}[x]$

$$P(x) = x^4 + 1 = (x^2+1)^2 - 2x^2 = (x^2+1)^2 - (\sqrt{2}x)^2$$

$$= \underbrace{(x^2+1-\sqrt{2}x)}_{\Delta < 0} \underbrace{(x^2+1+\sqrt{2}x)}_{\Delta < 0}$$

dans $\mathbb{C}[x]$.

$$x^n = 2 \in \mathbb{C}$$

$$= r e^{i\theta} \Rightarrow x = \sqrt[n]{r} e^{\frac{i\theta + 2ik\pi}{n}}$$

$$k = 0, 1, \dots, n-1$$

$$x^4 = -1 = e^{i\pi}$$

$$\Rightarrow x_k = 1 \cdot e^{\frac{i\pi + 2ik\pi}{4}}$$

$$k = 0, 1, 2, 3$$

$$\Rightarrow x_1 = e^{i\frac{\pi}{4}}; x_2 = e^{i\frac{3\pi}{4}}$$

$$x_3 = e^{i\frac{5\pi}{4}}; x_4 = e^{i\frac{7\pi}{4}}$$

... : d

2) $P(x) = x^4 + x^3 + 2x^2 + x + 1$.
dans $\mathbb{R}[x]$.

$$P(x) = (x^2+1)^2 + x(x^2+1) = (x^2+1)(x^2+1+x)$$

Dans $\mathbb{C}[x]$.

$$P(x) = (x^2+1)(x^2+x+1) = (x-i)(x+i)(x-j)(x-\bar{j})$$

avec $j = e^{i\frac{2\pi}{3}} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$.

3) $P(x) = x^4 - 2x^3 + 2x^2 - 2x + 1$.

$$= (x^2+1)^2 - 2x(x^2+1)$$

$$= (x^2+1)(x^2+1-2x)$$

$$= (x^2+1)(x-1)^2 \text{ dans } \mathbb{R}[x].$$

dans $\mathbb{C}[x]$

$$P(x) = (x-i)(x+i)(x-1)^2$$

Exercice 4:

$$P(x) = x^5 + 5x^4 + 11x^3 + 13x^2 + 8x + 2$$

$$P(-1) = -1 + 5 - 11 + 13 - 8 + 2 = 0$$

donc -1 est une racine de P

$$P'(x) = 5x^4 + 20x^3 + 33x^2 + 26x + 8$$

$$P'(-1) = 5 - 20 + 33 - 26 + 8 = 0$$

$$P''(x) = 20x^3 + 60x^2 + 66x + 26$$

$$P''(-1) = 86 - 86 = 0$$

$$P'''(x) = 60x^2 + 120x + 66$$

$$P'''(-1) = 126 - 120 = 6 \neq 0$$

donc -1 est une racine triple de P alors

$$P(x) = (x+1)^3 Q(x) \quad (\text{div euc})$$

$$= (x+1)^3 (x^2+2x+2) \text{ dans } \mathbb{R}[x].$$

$$\Delta = 4 - 8 = -4 < 0.$$

Exercice 5

1) $P(x) = x^4 + x^2 + 1$.

$$= x^4 + 2x^2 + 1 - x^2$$

$$= (x^2+1)^2 - x^2$$

$$= (x^2+x+1)(x^2-x+1) \text{ dans } \mathbb{R}[x]$$

dans $\mathbb{C}[x]$.

$$P(x) = (x-j)(x-\bar{j})(x+j)(x+\bar{j})$$

Rq:

j et $\bar{j}$ sont les racines de (x^2+x+1)

$-j$ et $-\bar{j}$ sont les racines de (x^2-x+1)

Exercice 6

$$B(x) = x^5 - 2x^4 + 2x^3 - 2x^2 + x$$

$$B(1) = 4 - 4 = 0$$

$\Rightarrow$ donc 1 est une racine de B .

$$B'(x) = 5x^4 - 8x^3 + 6x^2 - 4x + 1$$

$$B'(1) = 12 - 12 = 0$$

$$B''(x) = 20x^3 - 24x^2 + 12x - 4$$

$$B''(1) = 20 - 24 + 12 - 4 = 4 \neq 0$$

donc 1 est une racine double de B

$$B(x) = (x-1)^2 Q(x) \quad (Q(1) \neq 0)$$

2) $B(0) = 0$; $B'(0) = 1 \neq 0$

$\Rightarrow$ donc 0 est une racine simple de B

$$B(x) = x(x-1)^2 R(x) \quad (R(1) \neq 0 ; R(0) \neq 0)$$

3) $x^5 - 2x^4 + 2x^3 + x = (x^3 - 2x^2 + x)(x^2+1)$
 $= x(x^2+1)(x-1)^2$
 dans $\mathbb{R}[x]$.

4)
$$\begin{array}{r|l} x^5 - 2x^4 + 2x^3 - 2x^2 + x + 1 & B(x) \\ x^5 - 2x^4 + 2x^3 - 2x^2 + x & \\ \hline 1 & x \end{array}$$

$$A(x) = x B(x) + 1.$$

dans $\mathbb{R}(x)$.

Exercise 4:

$$P(x) = (x+1)^3(x^2+2x+2)$$

3) on pose $F(x) = \frac{1}{P(x)} = \frac{1}{(x+1)^3(x^2+2x+2)}$

$$= \frac{a}{x+1} + \frac{b}{(x+1)^2} + \frac{c}{(x+1)^3} + \frac{dx+e}{x^2+2x+2}$$

$$(x+1)^3 F(x) \Big|_{x=-1} = c = \frac{1}{x^2+2x+2} \Big|_{x=-1} \Rightarrow \boxed{c=1}$$

$$(x^2+2x+2) F(x) \Big|_{x=i-1} = d(i-1) + e = \frac{1}{(i-1+1)^3} = i$$

$$\Rightarrow e-d+id=i$$

$$\Rightarrow \begin{cases} e-d=0 \\ d=1 \\ e=1 \end{cases}$$

$$F(0) = a+b+c+\frac{e}{2} = \frac{1}{2}$$

$$\Rightarrow a+b+1=0$$

$$\Rightarrow a+b=-1$$

$$x F(x) \Big|_{x \rightarrow +\infty} = 0+a+d \Rightarrow \boxed{a=-1} \\ \Rightarrow \boxed{b=0}$$

$$\Rightarrow F(x) = \frac{1}{P(x)} = \frac{-1}{x+1} + \frac{1}{(x+1)^3} + \frac{x+1}{x^2+2x+2}$$

Exercise 5:

2) $F(x) = \frac{x^4}{P(x)} = \frac{x^4}{x^4+x^2+1}$

$$\frac{x^4}{-x^2-1} \Big| \frac{x^4+x^2+1}{1}$$

$$\Rightarrow F(x) = 1 + \frac{-x^2-1}{(x^2-x+1)(x^2+x+1)}$$

$$= 1 - \frac{x^2+1}{(x^2-x+1)(x^2+x+1)}$$

$$= 1 - H(x)$$

$$\Rightarrow H(x) = \frac{ax+b}{x^2-x+1} + \frac{cx+d}{x^2+x+1}$$

Rq: H pair alors $\begin{cases} a=-c \\ b=d \end{cases}$

1) $P(x) = x^4+x^2+1$

$$(x^2+x+1) H(x) \Big|_{x=j} = \frac{j^2+1}{j^2-j+1} = cj+d$$

$$\text{or } j^2+j+1=0 \Rightarrow j^2+1=-j$$

$$\Rightarrow \frac{j^2+1}{-2j} = cj+d = \frac{-j}{-2j} = \frac{1}{2}$$

$$\Rightarrow \boxed{d=\frac{1}{2}} \quad \boxed{c=0}$$

$$\Rightarrow \boxed{b=\frac{1}{2}} \quad \boxed{a=0}$$

$$\Rightarrow H(x) = \frac{1}{2(x^2-x+1)} + \frac{1}{2(x^2+x+1)}$$

$$\Rightarrow \frac{x^4}{P(x)} = 1 - \left[\frac{1}{2(x^2+x+1)} + \frac{1}{2(x^2-x+1)} \right]$$

$$\frac{1}{x^2+x+1} = \frac{1}{(x-j)(x-\bar{j})}; \frac{1}{x^2-x+1} = \frac{1}{(x+j)(x+\bar{j})}$$

$$\frac{1}{x^2+x+1} = \frac{a}{x-j} + \frac{b}{x-\bar{j}}$$

$$\frac{(x-\bar{j})}{(x-j)(x-\bar{j})} \Big|_{x=j} = a = \frac{1}{j-\bar{j}} = \frac{1}{2i \operatorname{Im}(j)}$$

$$\Rightarrow a = \frac{1}{2i \frac{\sqrt{3}}{2}} = \frac{1}{i\sqrt{3}} = \boxed{\frac{-i}{\sqrt{3}}}$$

$$\Rightarrow \boxed{b = \frac{i}{\sqrt{3}}}$$

$$\frac{1}{x^2-x+1} = \frac{1}{(x+j)(x+\bar{j})} = \frac{a}{x+j} + \frac{b}{x+\bar{j}}$$

$$\frac{x+j}{(x+j)(x+\bar{j})} \Big|_{x=-j} = a = \frac{1}{\bar{j}-j}$$

$$\Rightarrow a = \frac{1}{-2i \operatorname{Im}(j)} = \frac{1}{-2i \frac{\sqrt{3}}{2}} = \frac{1}{-i\sqrt{3}} = \boxed{\frac{i}{\sqrt{3}}}$$

$$\Rightarrow \boxed{b = \frac{-i}{\sqrt{3}}}$$

$$\Rightarrow \frac{x^4}{P(x)} = 1 - \frac{1}{2} \left(\frac{-i}{\sqrt{3}(x-j)} + \frac{i}{\sqrt{3}(x-\bar{j})} - \frac{i}{\sqrt{3}(x+j)} + \frac{i}{\sqrt{3}(x+\bar{j})} \right)$$

Exercise 6:

$$F(x) = \frac{1}{x(x-1)^2(x^2+1)} = \frac{a}{x} + \frac{b}{x-1} + \frac{c}{(x-1)^2} + \frac{dx+e}{x^2+1}$$

$$x F(x) \Big|_{x=0} = \boxed{a = 1}$$

$$(x-1)^2 F(x) \Big|_{x=1} = \boxed{c = \frac{1}{2}}$$

$$(x^2+1) F(x) \Big|_{x=i} = \frac{1}{i(i-1)^2} = di+e = \frac{1}{2} \Rightarrow \begin{cases} e = \frac{1}{2} \\ d = 0 \end{cases}$$

$$x F(x) \Big|_{x \rightarrow +\infty} = a + b + d = 0$$

$$\Rightarrow b = -a - d$$

$$\Rightarrow \boxed{b = -1}$$

$$\frac{1}{B(x)} = \frac{1}{x} - \frac{1}{x-1} + \frac{1}{2(x-1)^2} + \frac{1}{2(x^2+1)}$$

$$\frac{A(x)}{B(x)} = x + \frac{1}{B(x)}$$

Exercise 7:

$$F(x) = \frac{P(x)}{Q(x)} = \frac{(x-1)(x+1)(x-2)(x-j)(x-\bar{j})}{(x-2)^4 (x-1)(x+1)}$$

$$= \frac{(x-j)(x-\bar{j})}{(x-2)^3} = \frac{x^2+x-1}{(x-2)^3}$$

$$= \frac{a}{x-2} + \frac{b}{(x-2)^2} + \frac{c}{(x-2)^3}$$

$$(x-2)^3 F(x) \Big|_{x=2} \Rightarrow \boxed{c = 7}$$

$$x F(x) \Big|_{x \rightarrow +\infty} \Rightarrow \boxed{a = 1}$$